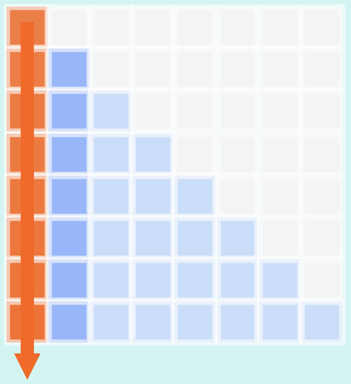
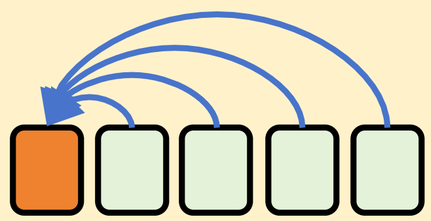
Attention sink



Value-path gradient aggregation

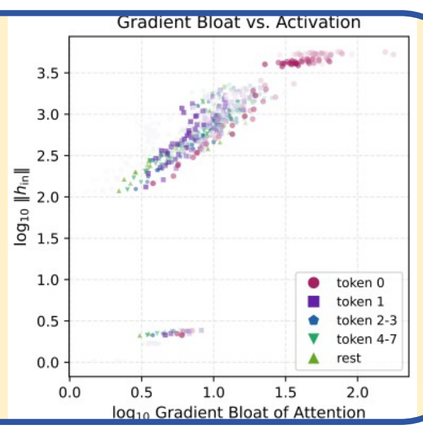


V-scale gradient valve

$$\hat{v} = \frac{\|v\|_2^2}{\|v\|_2^2 + C} v$$

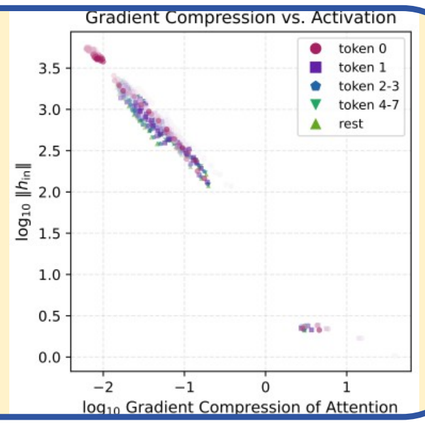
Attenuates value gradients at sink

Localized gradient pressure

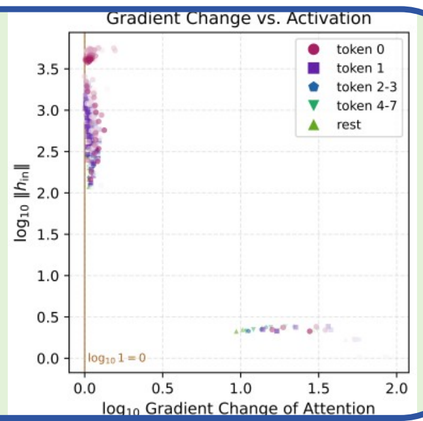


Reduced localized gradient pressure

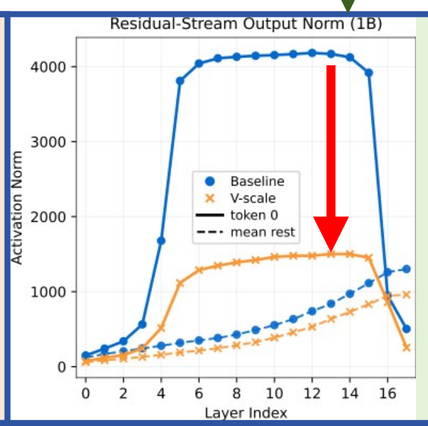
RMSNorm-mediated compression



Small gradient norm change



Massive activation



Suppressed massive activations